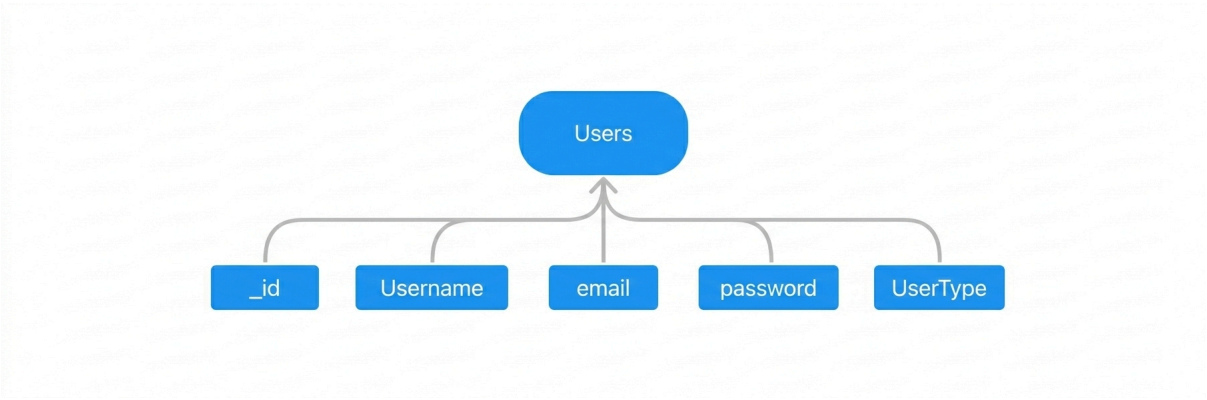
ER DIAGRAM

1.Users Schema:-



The **Users** entity stores the **information of all registered users** on the platform. It is responsible for managing **user authentication, identification, and access control**. This entity connects users with other parts of the system such as **portfolios, transactions, and stock activities**, allowing the platform to track each user's actions and provide a **personalized dashboard and secure login system**.

Attributes of Users Entity

• **_id:** The **_id** is the **primary key** that uniquely identifies each user in the database. It is usually **automatically generated** and helps link user records with other tables like **transactions and portfolios**.

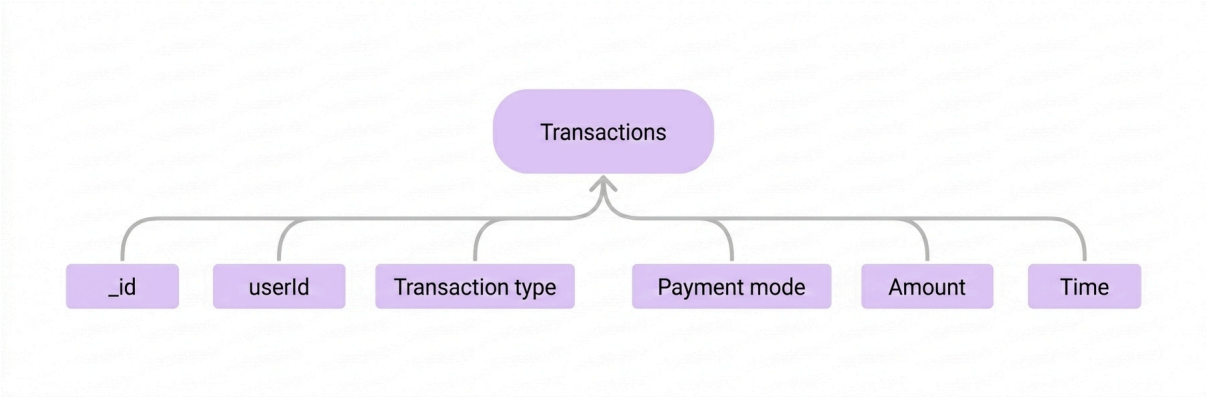
• **Username:** The **Username** is the **unique name selected by the user during registration**. It is used to **identify and display the user** within the platform.

• **Email:** The **Email** stores the **registered email address of the user**. It is used for **account verification, communication, and password recovery**.

• **Password:** The **Password** contains the **securely stored (hashed) login credential** used for **authentication and account security**.

• **UserType:** The **UserType** defines the **role of the user** in the system (such as **Admin or Trader**). It helps implement **role-based access control** and determines what features the user can access.

2.TransactionSchema:-



The **Transactions** entity records all **financial activities** performed by users on the platform. It maintains a complete record of **buy, sell, deposit, or withdrawal operations**, ensuring **transparency, tracking, and accurate financial management**. Each transaction is linked to a specific user and stores important information such as **transaction type, payment mode, amount, and time**, allowing the system to maintain a **complete transaction history** and monitor **trading activities efficiently**.

• **_id:** The **unique primary key** that **identifies each transaction record** in the database.

• **userId:** Stores the **unique identifier of the user** who performed the transaction and **links the transaction to the Users entity**.

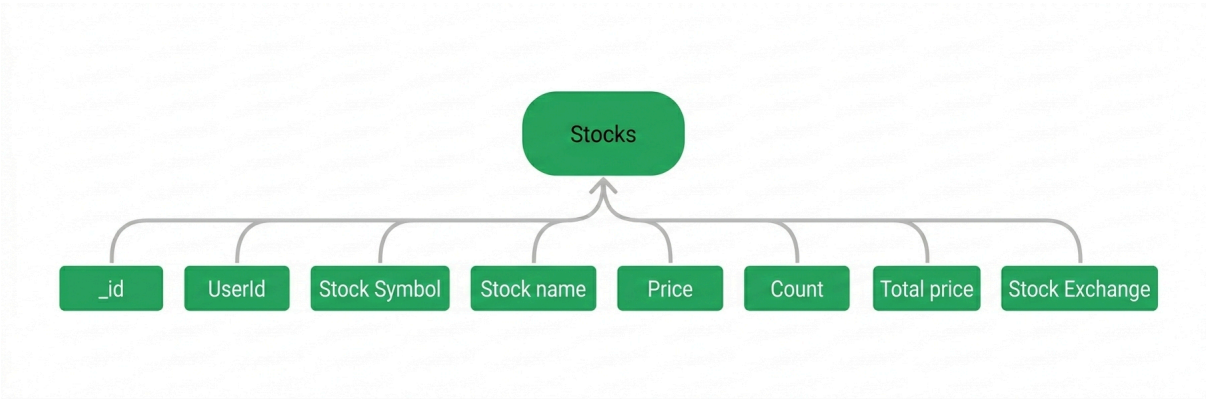
• **Transaction Type:** Defines the **type of financial operation**, such as **Buy, Sell, Deposit, or Withdrawal**.

• **Payment Mode:** Indicates the **method used to complete the transaction**, such as **wallet balance, virtual funds, or bank transfer**.

• **Amount:** Represents the **total monetary value** involved in the transaction.

• **Time:** Stores the **exact date and time** when the transaction occurred, helping maintain a **chronological transaction history**.

3.Stocks Schema:-



The **Stocks** entity stores all **information** related to **stocks available** on the platform. It keeps track of details such as the **stock symbol**, **company name**, **stock price**, **quantity of shares**, and **stock exchange**. This entity helps the system manage stock **listings**, **trading operations**, and **portfolio tracking**. By maintaining this information, the platform can provide users with accurate stock data, trading opportunities, and real-time investment analysis.

• **_id:** The **unique primary key** that **identifies each stock record** in the database.It ensures every stock entry is uniquely stored and managed.

• **UserId: Unique identifier** of the user associated with the stock record.It links the stock data to the Users entity.

• **Stock Symbol: Unique trading code** used to **identify a stock** in the market.It helps users quickly search and trade specific stocks.

• **Stock Name:** Name of the **company** whose **stock is being traded**.It provides **clear identification** of the company behind the stock.

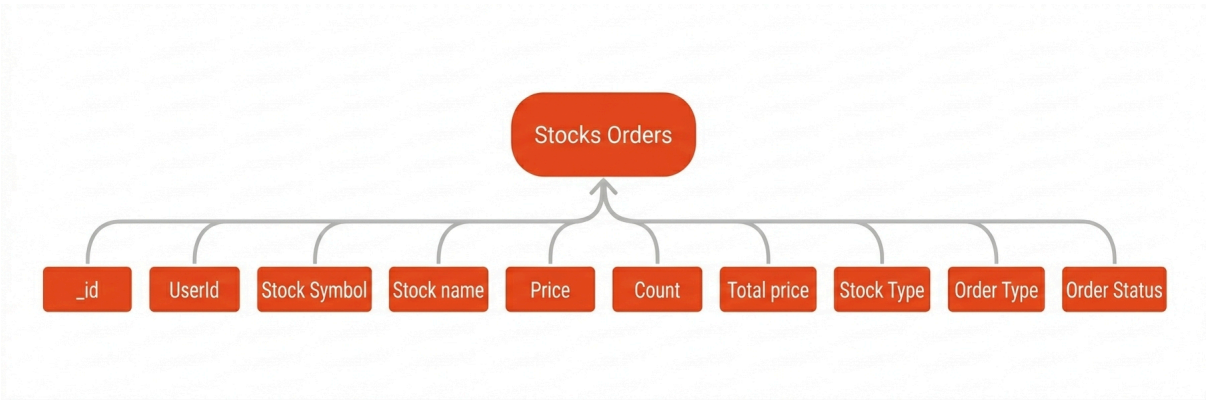
• **Price:** Current value of one share of the stock.It is used to **calculate investment value** and trading decisions.

• **Count:** Number of **shares owned or traded** by the user.It represents the quantity of stock in a **transaction** or **portfolio**.

• **Total Price:** Total value of shares calculated using **price × quantity**.It shows the overall **investment value** of the stock purchase.

• **Stock Exchange:** Market platform where the **stock** is **listed**.Examples include **NYSE** or **NASDAQ**, where trading takes place.

4.Stocks Orders Schema:-



The **Stock Orders entity** stores information about **all buy and sell orders** placed by users on the platform**.** It keeps track of details such as the **stock symbol, stock name, price, number of shares, order type, and order status.** This entity plays an important role in managing **stock trading activities, order processing, and portfolio updates.** By maintaining these records, the system can monitor **order execution, transaction accuracy, and user trading history, ensuring smooth and reliable stock trading operations.**

• **_id:** The **unique primary key** that **identifies each stock record** in the database.It ensures every stock entry is uniquely stored and managed.

• **UserId: Stores** the **unique identifier** of the user who placed the order and links it to the Users entity.

• **Stock Symbol:** Represents the **unique trading code** used to **identify** a stock in the **stock market.**

• **Stock Name:** Stores the name of the **company w**hose stock is being **ordered or traded.**

• **Count: Indicates the number of shares** included in the order.

• **Total Price:** Shows the **total value** of the order, calculated using **price × quantity** of shares.

• **Stock Type:** Defines the **category of stock,** such as equity or other trading types.

• **Order Type: Specifies** whether the order is a **Buy** order or **Sell** order.

• **Order Status:** Indicates the current state of the order, such as **pending, completed, or cancelled.**